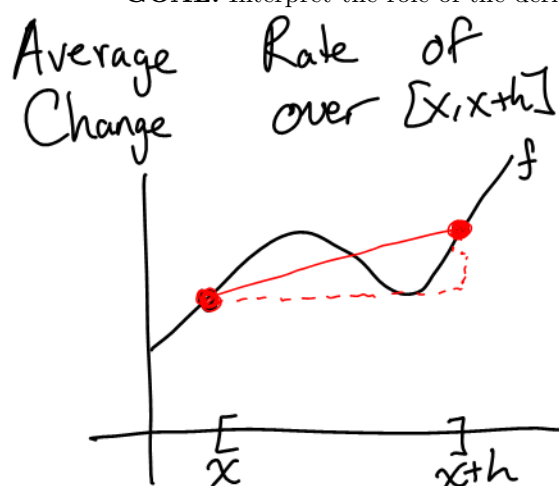


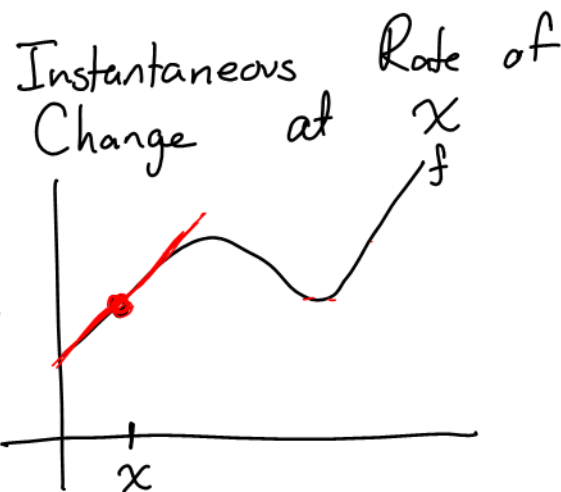
Lecture 7: Instantaneous Rates of Change

GOAL: Interpret the role of the derivative in various contexts.



$$\text{Avg Change} = \frac{f(x+h) - f(x)}{h}$$

$h \rightarrow 0$



$$\text{Inst. Change} = f'(x)$$

Physics

Velocity:

$$\frac{\text{Change in Position}}{\text{Time}} = \frac{\Delta s(t)}{\Delta t}$$

If $s(t)$ represents the position at time t , then the velocity $v(t)$ is given by

$$v(t) \stackrel{\text{def}}{=} \frac{ds}{dt}$$

★ Unless otherwise stated, t is time in seconds

Example 1

The position of a particle (in meters) moving in a straight line is given by

$$s(t) = 7 \cos t + t^2$$

$$\begin{aligned} \sin t^2 &\approx \sin(t^2) \\ \sin t^2 + 1 &= \sin(t^2) + 1 \end{aligned}$$

What's the velocity?

$$v(t) \stackrel{\text{def}}{=} \frac{ds}{dt} = [7 \cos t + t^2]' = 7 [\cos t]' + [t^2]'$$

$$= [7(-\sin t) + 2t] \frac{m}{s}$$

★ Unit of the Derivative = $\frac{\text{Unit of the Dependent Var}}{\text{Unit of the Independent Var}}$

Example 2

A ball is tossed straight into the air. The height from the ground (in feet) can be modeled by

$$s(t) = 80t - \underbrace{16t^2}_{-\frac{1}{2}g}$$

$$\begin{aligned} & -9.81 \frac{m}{s^2} \\ & 2g \\ & -32 \frac{ft}{s^2} \end{aligned}$$

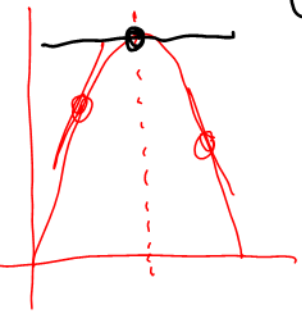
@ Velocity?

$$\begin{aligned} v(t) = s'(t) &= [80t]' - [16t^2]' \\ &= [80 - 32t] \frac{ft}{s} \\ &\quad 80[t]' = 80 \cdot 1 \end{aligned}$$

(b) When $t=3$, is the ball going up or down?

$$v(3) = 80 - 32 \cdot 3 = 80 - 96 = -16 \frac{ft}{s}$$

$v(3) < 0 \Rightarrow$ height is decreasing
 $\Rightarrow$ Ball is going down



(c) When will the ball reach its peak?

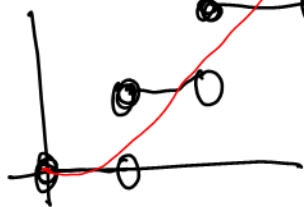
$$v(t) = 80 - 32t \stackrel{\text{set}}{=} 0$$

$$80 = 32t$$

$$t = \frac{80}{32} = 2.5 \text{ seconds}$$

Biology

Population Growth



$P(t)$ represents the population at time t

$$P(t) = P_0 \cdot e^{rt}$$

P_0 Initial Population

r Some Constant

$\frac{dP}{dt}$ is called the population growth; measures how the population is growing or shrinking

Example 3 The population of a city since 2000 can be modeled by

$$P(t) = 100t^2 - 600t + 10000$$

① What is the rate at which the population is changing?

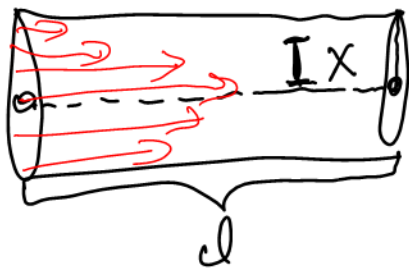
$$P'(t) = [200t - 600] \frac{\text{people}}{\text{year}}$$

② What year is the population increasing at a rate of 1000 people/yr

$$\begin{aligned} 200t - 600 &\stackrel{\text{set}}{=} 1000 \\ 200t &= 1600 \\ t &= 8 \end{aligned}$$

$\Rightarrow$ In the year 2008

Blood Flow The velocity of the blood is given



$$V(x) = \frac{\overset{\text{Pressure Difference}}{P}}{\underset{\text{Viscosity}}{4\eta} \underset{\text{Length}}{l}} (R^2 - x^2)$$

Example 4 In a small artery the velocity of the blood is given by

$$V(x) = (1.85 \times 10^4) (6.4 \times 10^{-5} - x^2)$$

x is in cm.

@ Find the velocity gradient, $V'(x)$

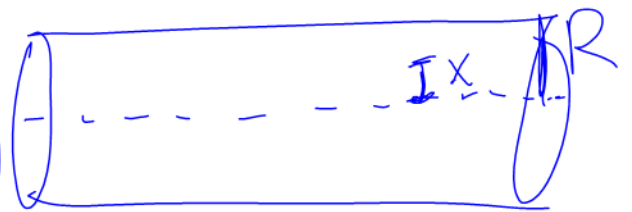
$$\begin{aligned} V'(x) &= (1.85 \times 10^4) [6.4 \times 10^{-5} - x^2]' \\ &= [-2(1.85 \times 10^4) x] \frac{\text{cm/s}}{\text{cm}} \end{aligned}$$

⑥ Evaluate at $x = 0.002 \text{ cm} = 2 \times 10^{-3} \text{ cm}$

$$\begin{aligned} V'(0.002) &= -2(1.85 \times 10^4)(2 \times 10^{-3}) \\ &= -74 \frac{\text{cm/s}}{\text{cm}} \end{aligned}$$

NOTE: $1 \text{ cm} = 10000 \mu\text{m}$

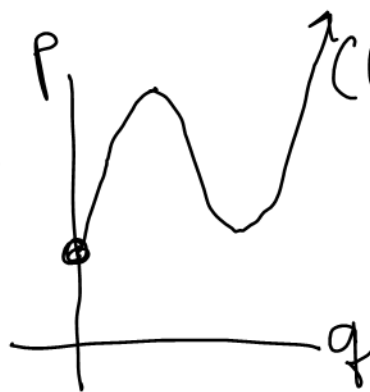
$$-74 \frac{\text{cm/s}}{\text{cm}} = -74 \left(\frac{\mu\text{m/s}}{\mu\text{m}} \right)$$



I.e., when we move $1 \mu\text{m}$ away from the center of the artery, the velocity decrease by $74 \frac{\mu\text{m}}{\text{s}}$

Economics

Cost Functions



The cost function $P = C(q)$ represents the cost to produce q units.

$C(0)$ is called the overhead costs.

$\frac{dP}{dq} = \frac{dC}{dq}$ is the marginal costs.

NOTE: It is common to model costs using a cubic function.

Example 5 In USD, a company estimates the cost of producing q units as

$$C(q) = 10000 + \underbrace{5q}_{\text{unit}} + 0.01q^3$$

What is the marginal costs at 50 items?

$$C'(q) = 5 + 0.01(3q^2) = 5 + 0.03q^2$$

$$C'(50) = 5 + 0.03(50)^2 = \$80/\text{item}$$

NOTE: $C'(50) \approx \frac{C(51) - C(50)}{51 - 50} = \$81.51/\text{item}$

Profit

The Revenue Function $[R(q)]$ is the amount of money made after producing q units (w/o factoring in expenses)

Profit $\stackrel{\text{def}}{=}$ Revenue - Cost

$$P(q) \stackrel{\text{def}}{=} R(q) - C(q)$$

Example 6

$$R(q) = q$$

$$C(q) = 50 - 6q^2 + 4q^3$$

Find the marginal profit?

$$P'(q) = [R(q) - C(q)]'$$

$$= R'(q) - C'(q)$$

$$= 1 - [-12q + 12q^2]$$

$$= -12q^2 + 12q + 1$$